ModelIt!

Student Activity Pack

Where Does Your Electricity Come From?

Tracing Energy from Natural Resources to Your Light Switch

Grade 4 | NGSS 4-ESS3-1 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What happens when you flip a light switch? Where does the electricity come from?

2 Can you name any ways that electricity is made?

3 What is the difference between energy from the sun and energy from coal?

4 Draw a picture showing how electricity gets from a power plant to your home.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Energy Source Amount | Power Plant Output | Home Electricity

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Sunny Day Solar

High energy source (lots of sunlight)

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Cloudy Week

Low energy source (very little sunlight)

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Windy Night

Moderate energy source (steady wind)

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Battery Storage — Batteries can store extra electricity from solar and wind for use when the sun is not shining or the wind is not blowing

Transmission Distance — Electricity loses some energy as heat when it travels long distances through power lines

Energy Demand — The amount of electricity people use changes throughout the day — highest in the evening when everyone is home

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

What happens between a power plant and your light switch to make electricity flow to your home? Use evidence from your simulation to support your answers.

1 Why can we not just use one type of energy source for everything?

2 What happens to a solar-powered town when it is cloudy for a whole week?

3 How does burning coal to make electricity affect the air and the environment?

4 If you could design the perfect energy source, what would it be like?

5 How does your model help you understand the trade-offs between different energy sources?

8 STEM Challenge: Design Your Town's Energy Plan

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YOUR ENGINEERING MISSION

Design an energy plan for a small town that provides reliable electricity while being as kind to the environment as possible.

THE SCENARIO

Your town of 1,000 homes needs a new energy plan. The town council wants you to choose the best mix of energy sources. You have options: a coal plant, a solar farm, a wind farm, or a small dam. Each has advantages and disadvantages. Use your model to recommend the best plan.

Engineering Design Thinking

1. WHAT MIX OF ENERGY SOURCES WILL YOU RECOMMEND FOR THE TOWN?
2. HOW WILL YOU MAKE SURE THERE IS ELECTRICITY AT NIGHT WHEN SOLAR PANELS DO NOT WORK?
3. WHAT ARE THE ENVIRONMENTAL TRADE-OFFS OF EACH ENERGY SOURCE YOU CHOSE?
4. HOW WILL YOUR PLAN CHANGE IF THE TOWN GROWS TO 2,000 HOMES?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Energy Engineers and Sustainability Planners design power systems and help communities switch to cleaner energy sources. They earn $65,000–$120,000/year.